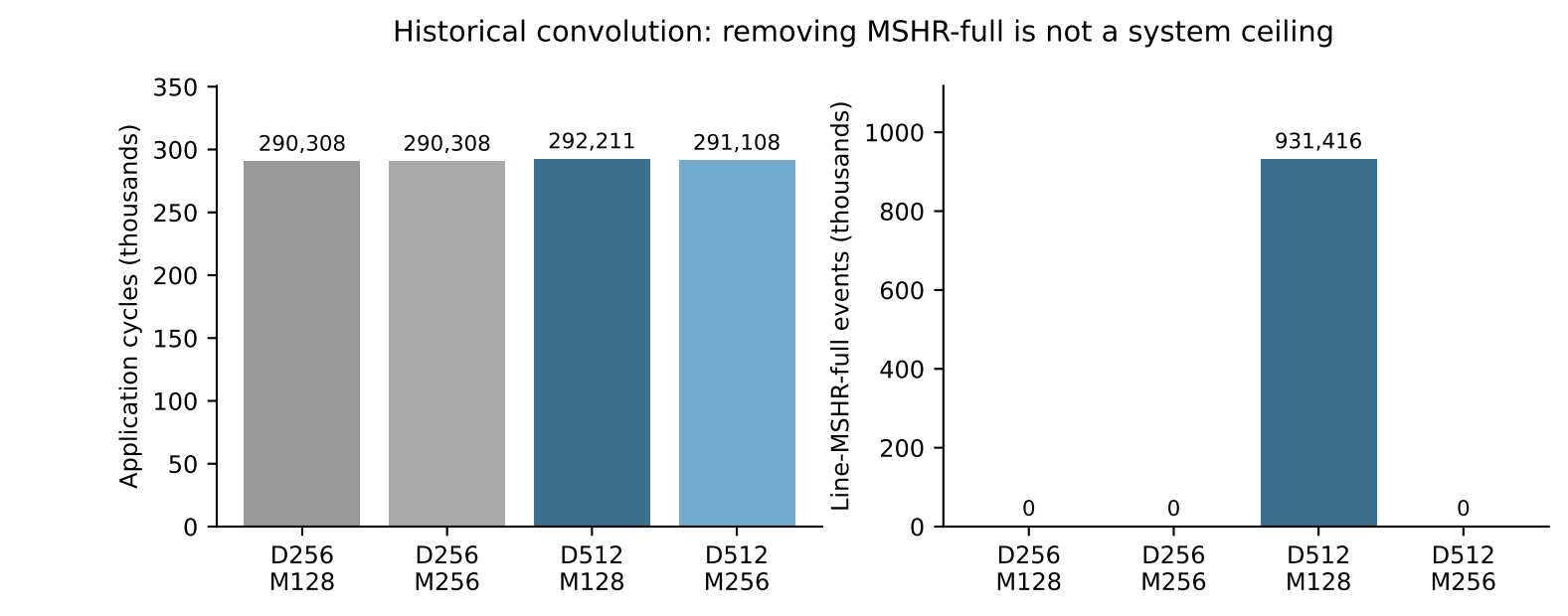

HISTORICAL_CALIBRATION: D512 M128→M256 gives 292211 / 291108 = 1.003789×; lower-path constraints remain.